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# Effect of aerobic pre-treatment on hydrogen and methane production in a two-stage anaerobic digestion process using food waste with different compositions

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ABSTRACT

Aerobic pre-treatment was applied prior to two-stage anaerobic digestion process. Three different food wastes samples, namely carbohydrate rich, protein rich and lipid rich, were prepared as substrates. Effect of aerobic pre-treatment on hydrogen and methane production was studied. Pre-aeration of substrates showed no positive impact on hydrogen production in the first stage. All three categories of pre-aerated food wastes produced less hydrogen compared to samples without pre-aeration. In the second stage, methane production increased for aerated protein rich and carbohydrate rich samples. In addition, the lag phase for carbohydrate rich substrate was shorter for aerated samples. Aerated protein rich substrate yielded the best results among substrates for methane production, with a cumulative production of approximately 351 ml/gVS. With regard to non-aerated substrates, lipid rich was the best substrate for $CH_4$ production (263 ml/gVS). Pre-aerated P substrate was the best in terms of total energy generation which amounted to 9.64 kJ/gVS. This study revealed aerobic pre-treatment to be a promising option for use in achieving enhanced substrate conversion efficiencies and $CH_4$ production in a two-stage AD process, particularly when the substrate contains high amounts of proteins.

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## 1. Introduction

The use of renewable energy sources is a critical issue worldwide due to the serious negative environmental consequences caused by the use of fossil fuels, in addition to the proximate depletion of the latter in the near future. Anaerobic digestion (AD) is one of the most widely investigated methods used in the production of energy from different kinds of organic waste. During this process, strictly anaerobic bacteria and archaea are utilized to produce biofuels such as hydrogen and methane when growing on organic substrates. Hydrogen has been indicated as one of the most promising fuels for the future (Ozkan et al., 2010; De Gioannis et al., 2013). However, subsequent to anaerobic hydrogen production substrate conversion remains incomplete, with the majority remaining as a residue after the process. A promising system is represented by a two-stage AD process combining $H_2$ and $CH_4$ productions. During the first stage, organic compounds are hydrolysed and utilized by hydrogen producing bacteria to produce $H_2$ and volatile fatty acids (VFAs), whilst in the second stage, VFAs are used as substrates for $CH_4$ production by methanogens. Two-stage AD provides a positive energy yield (40–90% available energy), thus underlining the highly important process sustainability (Ruggeri et al., 2010). Several studies have demonstrated the ability of two-stage AD to improve $CH_4$ yields during the second stage, likely due to better hydrolysis (Liu et al., 2006; Pakarinen et al., 2011). Moreover, compared to one-stage AD, process control would be simpler and stability would be improved (Lim and Wang, 2013; Ariunbaatar et al., 2015).

During hydrolysis, the rate limiting step of anaerobic digestion, organic compounds including proteins, carbohydrates and lipids are broken down by hydrolytic bacteria into amino-acids, sugars and long chain fatty acids, respectively. Substrate pre-treatment methods are aimed at promoting and improving hydrolysis of high molecular weight compounds to readily-biodegradable constituents, and subsequently increasing the AD process product yields.

Hydrolysis occurs under both aerobic and anaerobic conditions; however, hydrolysis rates are significantly higher under aerobic conditions, likely due to the higher production of enzymes (Botheju et al., 2009). In addition, pre-aeration reduces accumulation

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of VFAs, resulting in a drop of pH during the process, thus improving the start-up stability of anaerobic digestion. Limited pre-aeration prior to anaerobic digestion has been shown to improve hydrolysis and biogas production (Charles et al., 2009; Zhu et al., 2009; Ahn et al., 2014; Cossu et al., 2016; Peces et al., 2016).

Composition of organic wastes varies according to the source from which the wastes are collected. Slaughterhouse wastes may be rich in proteins and lipids, while food wastes and organic fraction of municipal solid wastes (OFMSW) are rich in carbohydrates. An in-depth understanding of effective pre-treatment methods for each kind of waste is fundamental in improving biogas production.

To the best of the Authors' knowledge, no scientific reports have been published to date on the effects of aerobic pre-treatment on food waste with different compositions for either $H_2$ and/or CH4 production in a two-stage AD process. Moreover, the effect of carbohydrate, lipid and protein content of food waste on pre-aeration efficiencies has not been addressed before. Therefore, the present work aims to study the aerobic pre-treatment effect of carbohydrate rich (C), protein rich (P), and lipid rich (L) food waste prior to two-stage anaerobic digestion on both $H_2$ and $CH_4$ production.

## 2. Materials and methods

### 2.1. Organic waste samples

Synthetic food waste samples were prepared in order to simulate industrial or municipal food waste with different compositions as indicated in a previous study (Alibardi and Cossu, 2016).

Three different substrates were prepared and classified as C, P, and L substrates. The composition of samples is shown in Table 1. The percentages are based on wet weight.

Food waste samples were shredded after preparation and characterized (Table 2) in order to have more detailed information for each substrate category.

### 2.2. Aerobic pre-treatment of the substrates

In order to compare the two-stage AD process with and without pre-aeration on the prepared substrates featuring different compositions, half the waste samples from each category were air injected using an aquarium pump (EIN WELTWEIT-Elite799) connected to a porous stone for better air diffusion. The air flow rate was fixed at 5 L/h using a flow meter (BROOKS SHO-RATE 1355). After 24 h, aeration was stopped. The inoculum was then added to each bottle with and without pre-treatment.

### 2.3. Two-stage AD – Hydrogen production

Laboratory scale tests were performed to evaluate Biochemical Hydrogen Potential (BHP) of the examined substrates. Batch tests were carried out using 1-L glass bottles which were subsequently sealed with silicon plug. Substrate concentration and food to microorganism ratio (F/M) were 5 gVS/l and 0.3 gVS/gVS, respectively. Granular sludge was used as inoculum for BHP and was collected from a full-scale Upflow Anaerobic Sludge Blanket (UASB) digester of a brewery factory located in Padova, Italy.

Heat treatment was carried out on granular sludge in a rotary water-bath incubator at a fixed temperature of 80 °C for 15 min in order to suppress methanogenic bacteria (Alibardi and Cossu, 2016). pH was set at 6.0 using phosphate buffer before the start of tests. Three main $H_2$-producing enzymes are used by anaerobic microorganisms: [Fe/Fe]/hydrogenases, [Ni/Fe]/hydrogenases and nitrogenases. These $H_2$-producing enzymes are generally all highly oxygen-sensitive and presence of oxygen may reduce their activities (Mathews and Wang, 2009). Accordingly, $H_2$ production should be carried out under strictly anaerobic conditions. Therefore, following aerobic pre-treatment, the bottles were flushed with N2 gas for 3 min to ensure anaerobic conditions and incubated at a temperature of 35 ± 1 °C. All tests were performed in duplicate.

**Table 1**
Composition of synthetic food wastes (%W/W).

| Ingredients | C | L | P |
|---|---|---|---|
| Tuna (%) | 6.7 | 7.5 | 31.1 |
| Butter (%) | 5.5 | 22.3 | 5.5 |
| Apple (%) | 27.8 | 27 | 7.85 |
| Banana (%) | 27.8 | 27 | 7.85 |
| Chicken breast (%) | 6.7 | 7.5 | 31.1 |
| Bread (%) | 5.4 | 1.5 | 3.2 |
| Pasta (%) | 5.4 | 1.5 | 3.2 |
| Minestrone soup (%) | 14.7 | 5.5 | 10.2 |

**Table 2**
Average characteristics of food wastes with different compositions.

| Parameters | C | L | P |
|---|---|---|---|
| TS (%) | 28.56 | 30.72 | 43.2 |
| VS (%TS) | 95.4 | 96.1 | 97.3 |
| TOC (%TS) | 58.7 | 65.9 | 66.3 |
| TKN (%TS) | 3.34 | 3.05 | 7.98 |
| Lipid (%TS) | 16.1 | 41 | 17.3 |
| Protein (%TS) | 19.8 | 18.1 | 47.3 |
| Glucose (%TS) | 4.2 | 1.54 | 3.11 |
| Fructose (%TS) | 12.36 | 5.29 | 2.75 |
| Sucrose (%TS) | 15.56 | 7.42 | 2.78 |

### 2.4. Two-stage AD – Methane production

After completing the $H_2$ production phase, the bottles were opened and pH, dissolved organic carbon (DOC) and VFAs were measured. Non-pre-treated granular sludge (at the same amount as the first stage) was then added to each bottle and all were sealed again, flushed with $N_2$ gas for 3 min, and incubated at the same initial mesophilic conditions of 35 ± 1 °C.

### 2.5. Analytical methods

Total solids (TS), volatile solids (VS), and TKN were analysed according to standard methods (APHA, 1999). Total organic carbon (TOC) values were calculated on the basis of the difference between total carbon and inorganic carbon present in the samples. Concentrations of carbohydrates, proteins, lipids and free sugars were obtained according to official methods (AOAC, 2003). The volume of biogas produced during the anaerobic digestion process was measured by means of the water displacement method. The produced gas composition in terms of $H_2$ first, and then $CH_4$, was analysed using a micro-GC (Varian 490-GC) equipped with an MS5A column to measure $H_2$ and $CH_4$, and a PPU column for $CO_2$ and two Thermal Conductivity Detectors. Argon was used as the carrier gas at a pressure of 60 kPa. Temperatures of column and injector were set to 80 °C.

VFAs concentrations were measured using a gas chromatograph (Varian 3900) equipped with a CP-WAX 58 WCOT fused silica column and a Flame Ionization Detector. Nitrogen was used as carrier gas with a flow of 4 ml/min in column. The oven temperature was set at 80 °C for the first minute and then increased at a rate of 10 °C/min to 180 °C for two minutes. Column and injector temperatures were set to 250 °C.

### 2.6. Hydrogen and methane production calculations

Hydrogen, methane and carbon dioxide volumes produced during the first and second stages of AD were calculated according to Eq. (1) (Van Ginkel et al., 2005):

$$V_{c,t} = C_{c,t} * V_{b,t} + V_H * (C_{c,t} - C_{c,t-1}) \quad (1)$$

In which:

$V_{c,t}$ = Volume of $H_2$, $CH_4$ or $CO_2$ produced between intervals of t and t−1
$V_{b,t}$ = Volume of total biogas produced between intervals of t and t−1
$V_H$ = Volume of headspace of bottles
$C_{c,t}$ = Concentrations of $H_2$, $CH_4$ or $CO_2$ measured at time t
$C_{c,t-1}$ = Concentrations of $H_2$, $CH_4$ or $CO_2$ measured at time t−1

## 3. Results and discussion

### 3.1. Effect of aeration pre-treatment on the first stage of AD

#### 3.1.1. Hydrogen production

The results obtained for hydrogen production potential from three different food waste samples are shown in Fig. 1. Data obtained through GC analysis revealed a lack of methane in the emitted gas, due to efficiency of the thermal pre-treatment of inoculum. In the first stage of AD, substrate C without aeration produced considerably more hydrogen (55.31 ml/gVS) compared to L (27.93 ml/gVS) and P (7.96 ml/gVS) substrates. This finding is in agreement with Alibardi and Cossu (2015), who concluded that carbohydrate rich food waste is capable of producing much higher quantities of $H_2$ compared to lipid or protein rich substrates. This could be attributed to faster hydrolysis rate of carbohydrates (almost 20 times faster) compared to lipids and proteins (Lay et al., 2003). Since the duration of $H_2$ production is short (around 3 days) it is not enough for the hydrolysis of proteins and lipids. In addition, conversion of long chain fatty acids from hydrolysis of lipids to $H_2$ is feasible only at very low hydrogen partial pressure (Hallenbeck, 2009). Degradation of some amino acids from hydrolysis of protein is $H_2$ consuming. According to Hallenbeck (2009), readily-biodegradable carbohydrates are the preferred substrates by anaerobic microorganisms during dark fermentative $H_2$ production. Similarly, Chu et al., (2012) showed that $H_2$ yield is strongly dependent on the carbohydrate content of organic wastes.

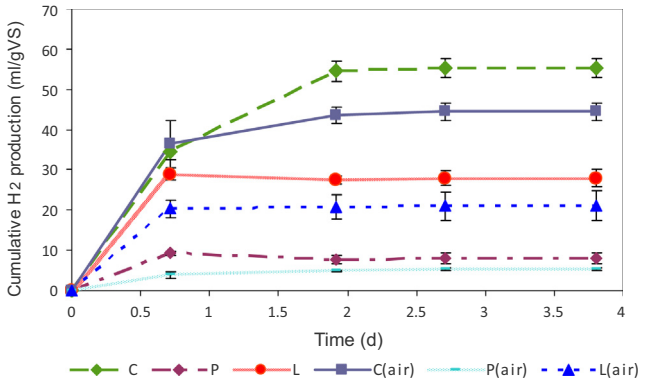



**Fig. 1.** Hydrogen production potential for the three different substrates, namely C, P, and L, with and without aerobic pre-treatment. C, P and L are carbohydrate, protein and lipid rich substrates without aerobic pre-treatment, respectively. C (air), P (air) and L (air) are carbohydrate, protein and lipid rich substrates with aerobic pre-treatment, respectively.

For samples subjected to aerobic pre-treatment, substrate C achieved the highest $H_2$ yield (44.4 ml/gVS), followed by substrate L (21 ml/gVS) and P (5.27 ml/gVS). Aerobic pre-treatment lowered average $H_2$ production for C (19%), L (24%), and P (33%) substrates. Although aerobic pre-treatment has been indicated as a strategy to increase hydrolysis and $CH_4$ production from sludge (Ahn et al., 2014; Peces et al., 2016), it proved ineffective in achieving an increase in $H_2$ production from food waste. The latter could convincingly be explained by a low solid retention time (SRT) for $H_2$ (3 days) compared to $CH_4$ production (15 days or more). In addition, during aeration, part of the available readily-biodegradable carbon (mainly free sugars) is converted to $CO_2$ or consumed for cell growth instead of in product formation (Botheju and Bakke, 2011). In the present study, the carbon loss after aeration was proved by 37%, 6%, and 12% decreased TOC content for C, P, and L substrates, respectively. Although this drawback may also be present in pre-aeration studies on $CH_4$ production, it may be compensated by a longer SRT, which enhances carbon hydrolysis with lower degradability, subsequently leading to higher product yields.

#### 3.1.2. VFA composition after the hydrogen-producing phase

The results of characterization of liquid samples at the end of the hydrogen-producing phase are shown in Tables 3 and 4. The major VFA components for all food waste samples were acetic acid (AC) and butyric acid (BU), while propionic acid (PA) was detected only in P-rich samples. Aerobic pre-treatment slightly increased PA production for substrate P. Presence of PA at the end of the first stage is not favourable since unlike AC and BU it is produced by a metabolic pathway which consumes substrate without producing $H_2$ (Hawkes et al., 2007). The average concentrations of VFAs at the end of the first stage of AD are reported in Table 3. When compared to non-aerated samples, AC content in pre-aerated P samples was slightly lower (4%) while in pre-aerated samples and L AC decrease was much more significant, 33% and 25%, respectively. Similarly, aeration lowered BU production for sample C by 43%. On the contrary, in samples P and L, BU concentration increased by 34% and 10%, respectively. PA concentration in pre-aerated P samples was slightly higher (5%) than in non-pre-treated P ones.

Samples P, both with and without aeration, displayed the lowest BU/AC ratios amongst all substrate types. For non-aerated samples, correlation between BU/AC ratio and $H_2$ production is in agreement with previous studies which suggested that BU/AC ratio is directly proportional to $H_2$ yield (Kim et al., 2006). Conversely, other studies reported the absence of a correlation between BU/AC ratio and $H_2$ yield (Guo et al., 2014; Ghimire et al., 2015). Indeed, Table 3 highlights how for aerated P and L samples the BU/AC ratios increased in comparison to non-aerated samples, although lacking any positive effect on $H_2$ production.

pH, DOC concentrations and cumulative $CO_2$ production values obtained following the first stage for samples with and without aerobic pre-treatment are shown in Table 4. Pre-aeration was not found to have significantly affected cumulative $CO_2$ production at the end of the first stage. However, DOC values were lower for all pre-aerated samples in comparison to samples without pre-aeration. This could be mainly due to lower amount of easily degradable carbon in pre-aerated samples as a result of partial loss of carbon during aeration.

pH values were measured at the end of the first stage for all samples. Substrate P (both with and without pre-aeration) showed higher values compared to other substrates. Generally, substrates with high nitrogen content (such as protein rich wastes) prevent excessive acidification due to their buffering capacity (Boni et al., 2013).

In terms of pH, during the first stage, aeration led to a slightly higher pH in substrate C as a consequence of the lower VFA

**Table 3**
Average volatile fatty acid (VFA) production for the three different substrates, namely C, P, and L, with and without pre-treatment.

| Specific VFA (mg/l) | C | C (air) | P | P (air) | L | L (air) |
|---|---|---|---|---|---|---|
| Acetic acid | 593 ± 66 | 392 ± 28 | 490 ± 63 | 473 ± 51 | 510 ± 43 | 381 ± 29 |
| Butyric acid | 413 ± 50 | 236 ± 23 | 139 ± 35 | 187 ± 32 | 220 ± 38 | 243 ± 43 |
| Propionic acid | 0 | 0 | 88.9 ± 11 | 94.2 ± 21 | 0 | 0 |
| BU/AC ratio (mmol/mmol) | 0.47 | 0.41 | 0.19 | 0.26 | 0.29 | 0.43 |
| Total VFAs (mg/l) | 1006 | 628 | 717.9 | 754.2 | 730 | 624 |

**Table 4**
Average $CO_2$ and DOC concentration and pH at the end of first and second stages of AD process for the three different substrates, namely C-rich, P-rich, and L-rich with and without aerobic pre-treatment.

| | Parameter | C | C (air) | P | P (air) | L | L (air) |
|---|---|---|---|---|---|---|---|
| First stage | $CO_2$ (ml/gVS) | 82.42 ± 2.11 | 72.5 ± 1.41 | 42.81 ± 2.02 | 39.41 ± 2.11 | 65.64 ± 3.87 | 65.41 ± 2.5 |
| | DOC (mg/l) | 1003.5 ± 23.11 | 626.5 ± 31.33 | 734.5 ± 2.08 | 690.5 ± 8.12 | 790.25 ± 28.14 | 678.5 ± 16.2 |
| | pH | 4.56 ± 0.01 | 4.88 ± 0.02 | 5.51 ± 0.1 | 5.31 ± 0.03 | 4.56 ± 0.1 | 4.56 ± 0.1 |
| Second stage | $CO_2$ (ml/gVS) | 169.76 ± 3.2 | 159.18 ± 2.65 | 150.16 ± 3.37 | 214.62 ± 4.8 | 151.94 ± 2.12 | 151.95 ± 3.19 |
| | DOC (mg/l) | <15 | <15 | <30 | <25 | <20 | <30 |
| | pH | 7.49 ± 0.03 | 7.39 ± 0.1 | 7.69 ± 0.02 | 7.56 ± 0.02 | 7.3 ± 0.07 | 7.14 ± 0.1 |

concentration (see Table 3). Instead, aeration led to a lower pH in substrate P compared to non-aerated samples. For substrate L, with and without aerobic pre-treatment, pH values at the end of the first stage of the AD process were substantially similar.

### 3.2. Effect of pre-aeration on the second stage of AD

#### 3.2.1. Methane production

Cumulative methane productions for the three investigated food waste substrates, with and without aeration, are shown in Fig. 2. Long lag phase of almost one week (except for non pre-aerated sample C where the lag phase lasted about 3 weeks) was observed for all substrate types. The most probable reason was a low pH following completion of the first AD stage. However, for substrate P without aeration the lag phase lasted only 3 days.

For substrate C, aerobic pre-treatment ensured a better acclimatization of bacteria and increased cumulative $CH_4$ production by 6% was observed at the end of second stage of AD. Cumulative $CH_4$ production for the aerated substrate C was 600% higher than the non-aerated substrate C at day 14. Significant $CH_4$ production in non pre-aerated samples C started after about three weeks. Similarly, Charles et al. (2009) observed an accelerated $CH_4$ production after aerating OFMSW, a carbohydrate rich substrate.

For substrate P with aeration, cumulative $CH_4$ production was lower compared to non-aerated samples until day 50. After this time, $CH_4$ production remained virtually constant for non-aerated samples, whilst it increased significantly (45.6%) for samples with pre-aeration. Cumulative $CH_4$ production for P-rich with and without aerobic pre-treatment was 351.69 and 241.52 ml/gVS, respectively.

Cumulative $CH_4$ productions for substrate L were around 263 and 240 ml/gVS for non-aerated and pre-aerated samples, respectively. L was the only substrate that produced less $CH_4$ with pre-aeration. However, when taking into consideration non-aerated substrates, L was the best substrate type for $CH_4$ production in two-stage AD. Similarly, Johansen and Bakke (2006) reported that micro aeration led to higher hydrolysis of carbohydrates and proteins of primary sludge, while lipids hydrolysis failed to increase without the addition of inoculum.

Alibardi and Cossu (2015) reported that proteins and lipids produce higher quantities of $CH_4$ than carbohydrates. These Authors separated different fractions of municipal solid waste to produce $CH_4$ in single stage AD from each single fraction. They observed that the highest $CH_4$ production was achieved using the fraction containing meat, cheese and fish, and the lowest was produced using a fraction containing bread and pasta alone. This finding is in agreement with the present study in which substrates P and L produced more $CH_4$ compared to substrate C.

Several researchers have observed higher $CH_4$ production in single stage AD process following aerobic pre-treatment. Lim and Wang (2013) showed that aerobic pre-treatment of a mixture of brown water and food waste improved AD treatment performance with a 10% increase in $CH_4$ production. Pre-aeration of sewage sludge and primary sludge increased cumulative $CH_4$ production by 25% (Ahn et al., 2014), and 14% (Peces et al., 2016), respectively. According to Botheju et al. (2009) pre-aeration may increase substrate conversion efficiency due to enhanced hydrolysis.

Jang et al., (2014) applied aerobic thermophilic pre-treatment prior to mesophilic AD for sludge digestion. They obtained higher $CH_4$ production and higher carbon conversion efficiencies with aeration. The study was reported as using sludge with a higher protein compared to carbohydrate content.

In anaerobic digesters VFA accumulation, and consequent drop in pH, is a major issue during $CH_4$ production. Limited aeration may remove excess degradable carbon, thus promoting the onset of methanogenesis. Several studies have reported lower VFA accumulation during anaerobic digestion following aerobic pre-treatment (Chu et al., 1994; Zhou et al., 2007). The present study likewise showed positive effects of pre-aeration on the reduction of VFA accumulation during the second stage of AD (Fig. 3). A high VFA concentration was observed for substrate C without aerobic pre-treatment; lower VFA concentrations were observed in pre-aerated C samples, as well as higher $CH_4$ production (Fig. 2a).

Table 4 illustrates pH, $CO_2$ and DOC values after the second stage of AD. At the end of the second stage, pH values for all samples ranged from 7.1 to 7.6. Similar to the first stage, pH was slightly lower in pre-aerated samples, possibly due to higher carbon conversion to VFAs. The most significant difference between cumulative $CO_2$ production for pre-aerated and non-aerated samples was observed for P-rich substrate (214.62 and 150.16 ml/gVS, respectively). This sample also showed the highest $CH_4$ production.

### 3.3. Total energy yields from $H_2$ and $CH_4$ in the two-stage AD

$H_2$ and $CH_4$ productions obtained with each substrate revealed that more $H_2$ were accompanied by a lower production of $CH_4$ and vice versa. In order to define the efficiency of the two-stage process

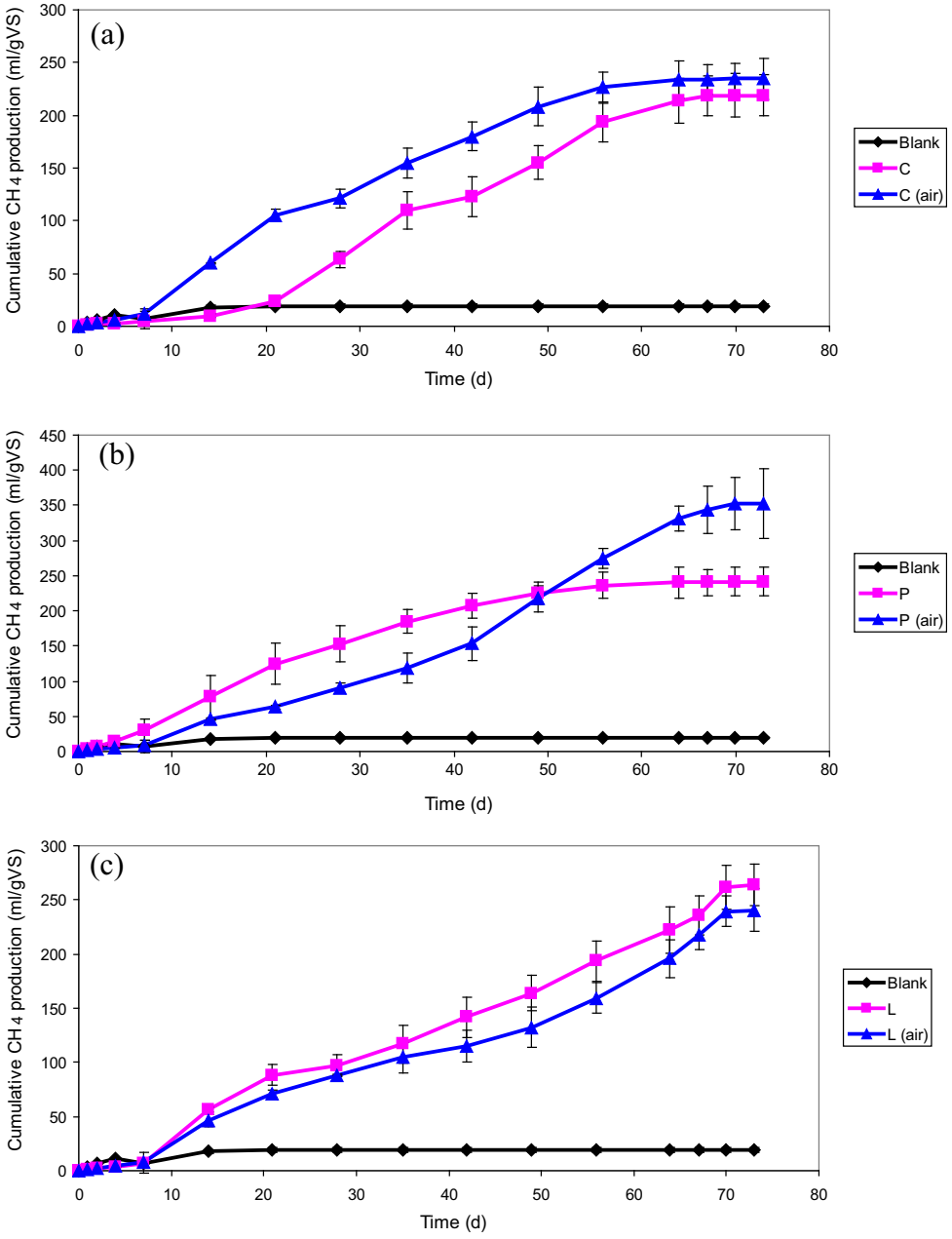



**Fig. 2.** Methane production potential of the three different substrates, namely C (a), P, (b) and L (c), with and without aerobic pre-treatment, after hydrogen production phase. C (air), P (air) and L (air) are carbohydrate, protein and lipid rich substrates with aerobic pre-treatment, respectively. C, P, L are carbohydrate, protein and lipid rich substrates without aerobic pre-treatment. B is blank.

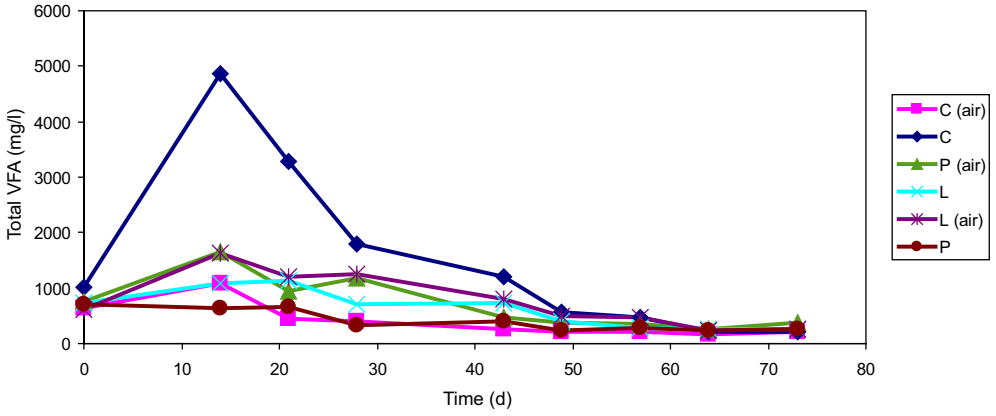



**Fig. 3.** VFA accumulation during the second stage of the AD process.

**Table 5**
Total energy generation in the process in terms of $H_2$ and $CH_4$.

| Substrate | Energy from $H_2$ (kJ/gVS) | Energy from $CH_4$ (kJ/gVS) | Total energy (kJ/gVS) |
|---|---|---|---|
| C | 0.23 | 5.98 | 6.21 |
| C (air) | 0.2 | 6.4 | 6.6 |
| P | 0.038 | 6.58 | 6.61 |
| P (air) | 0.017 | 9.63 | 9.64 |
| L | 0.14 | 7.22 | 7.36 |
| L (air) | 0.14 | 6.58 | 6.72 |

in terms of energy generation, total energies from $CH_4$ and $H_2$ have been calculated and presented in Table 5. Pre-aerated P substrate was the best in terms of total energy generation which amounted to 9.64 kJ/gVS, 45.8% higher than non pre-aerated P substrate. The increase of total energy yield was lower for pre-aerated C samples (6.3% higher than without aeration). Conversely, L samples performed better without aeration.

## 4. Conclusions

The efficiencies of a two-stage AD treatment using organic wastes with different compositions in both the presence and absence of aeration as a treatment were compared by evaluating the $H_2$ and $CH_4$ production. This study suggested that pre-aeration of organic waste did not constitute an effective treatment for the purpose of improving $H_2$ production potential during the first stage of the AD process. However, during the subsequent stage of AD, carbon conversion to $CH_4$ was higher for pre-aerated P and C samples than in non-pre-aerated ones, resulting in an increase of total energy potential. Conversely, pre-aeration of substrate L was not effective, causing a decrease of energy potential compared to non pre-aerated sample.

Further studies should however be undertaken using shorter and therefore less expensive, pre-aeration times in order to assess whether this may result in a positive effect on both $H_2$ and $CH_4$ productions. From the very beginning of the food waste treatment up to digestate management and disposal, the totality of the two-stage AD processes, with and without aeration, should be investigated and compared in terms of a complete Life Cycle Assessment, considering also the chemical and ecotoxicological quality of the produced digestate (Pivato et al., 2016a,b; Pivato et al., 2014).

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